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COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE:CSA 0611

SLOT:C

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CO4_AT2

1.Predict traffic congestion using historical data and Dynamic Programming techniques.

Sample Input: Historical Traffic Data.

Sample Output: Predicted Congestion Level

CODE:

```python
traffic = list(map(int, input("Enter traffic data: ").split()))
dp = [0] * len(traffic)
dp[0] = traffic[0]
 for i in range(1, len(traffic)):
    dp[i] = dp[i - 1] + traffic[i]
 average = dp[-1] / len(traffic)
if average >= 70:
   print("Predicted Congestion Level: High")
elif average >= 40:
   print("Predicted Congestion Level: Medium")
else:
   print("Predicted Congestion Level: Low")
```

OUTPUT:

```
Output

Enter traffic data: 20 40 60 80 100
Predicted Congestion Level: Medium

=== Code Execution Successful ===
```